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Practice guidelines

Diagnosis of Alzheimer's disease: Recommendations from the French Federation of Memory Clinics



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INFO ARTICLE

Article history:

Received 14 May 2025

Received in revised form

27 August 2025

Accepted 3 September 2025

Available online 30 September 2025

Keywords:

Alzheimer's disease

Diagnosis

Guidelines

Recommendations

ABSTRACT

Alzheimer's disease (AD) is the most common neurodegenerative disorder and the leading cause of major neurocognitive disorder in older adults. Its diagnosis has evolved from clinical to clinico-biological criteria, integrating biomarkers such as beta-amyloid and phosphorylated tau in cerebrospinal fluid or specific positron emission tomography (PET) imaging. Recent therapeutic advances, including anti-amyloid immunotherapies, highlight the need for early and accurate diagnosis. Clinical presentation is heterogeneous and may include amnesic or non-amnesic forms. Diagnosis should be suspected in patients with progressive cognitive decline and confirmed through neuropsychological assessment and biomarker testing. Blood-based biomarkers are promising but not yet validated for routine use. Magnetic resonance imaging (MRI) is recommended for all patients with recent cognitive decline. The role of general practitioners in early detection is critical. These recommendations, developed by the French federation of memory clinics, provide guidance

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<https://doi.org/10.1016/j.neurol.2025.09.004>

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AD biomarkers

on diagnosis stages, biomarker indications, first-line assessments, referral criteria, and communication of diagnosis. They aim to standardize clinical practice and support timely, individualized care.

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1. Introduction

Alzheimer's disease (AD) is the most common cause of cognitive decline [1], affecting approximately 1.2 million people in France [2] and 57 million worldwide [3]. As the global population ages, the number of individuals affected by AD is expected to rise significantly in the coming decades [4]. The disease has a profound impact on individuals, families, and society [5], being the leading cause of loss of autonomy in older adults and a major reason for admission to long-term care facilities [6].

The neuropathological hallmarks of AD are the accumulation of beta-amyloid plaques and neurofibrillary tangles composed of hyperphosphorylated tau protein [7]. These pathological changes are accompanied by synaptic loss, neuronal death, and gliosis. Historically, the diagnosis of AD was primarily based on clinical criteria [8], which were not highly specific, leading to about 30% false-positive diagnoses [9]. This is due to the clinical overlap with other conditions such as vascular cognitive disorders, dementia with Lewy bodies, frontotemporal lobar degeneration, psychiatric disorders [10], or the more recently described entity, limbic-predominant age-related TDP-43 encephalopathy (LATE) [11]. Recent advancements in the identification of specific biomarkers have significantly improved the accuracy of AD diagnosis [12]. These biomarkers include biochemical assays of beta-amyloid and tau proteins in cerebrospinal fluid (CSF) or in blood, when fully approved in clinical care, and imaging techniques like positron emission tomography (PET) scans using radioligands that bind to amyloid plaques or tau protein, the latter currently only being used for research purposes in France [13]. The integration of these biomarkers with clinical assessment has enabled a more precise diagnosis [14–16], even at the stage of mild cognitive impairment [17], where objective cognitive decline is present without impacting activities of daily living.

Despite the high prevalence of AD, its diagnosis remains challenging, with an estimated more than 50% of cases going undiagnosed [18]. The complexity of diagnosis is due to the chronic and progressive nature of the disease [19], its variable impact on cognitive function and daily activities [20], and the persistence of misconceptions about normal aging and neurocognitive disorders [21]. The limited efficacy of symptomatic treatments also hampers the early detection and referral of patients from primary care to specialized memory centers [22].

Over the past two decades, several diagnostic criteria have been proposed for AD, evolving to integrate clinical and biological markers. The NIA-AA criteria (2011) relied primarily on clinical assessments [23], distinguishing between amnesic

and non-amnesic phenotypes and using biomarkers and imaging as complementary tools. The NIA-AA Research Framework (2018) introduced a biologically-based approach [24], focusing on biomarkers for beta-amyloid, tau, and neurodegeneration, and is mainly used in research settings. The IWG-2 criteria in 2021 [25], reinforced and updated in 2024 [26], defined AD as a clinico-biological entity requiring both clinical symptoms commonly associated with AD neuropathological change and positive biomarkers for beta-amyloid and tau, addressing different clinical forms of the disease. The most recent criteria, updated by the Alzheimer's Association Workgroup in 2024 [27], emphasize a biological process beginning with asymptomatic neuropathological changes, with any core 1 biomarker equivalent to amyloid-PET being sufficient for diagnosis and guiding clinical decisions.

In 2023, two positive phase III studies on immunotherapies targeting beta-amyloid for early-stage Alzheimer's disease were published [28,29]. These studies showed that targeting beta-amyloid plaques may slow the progression of AD in its early stages, highlighting the importance of early diagnosis. The risk of amyloid-related imaging abnormalities (ARIA) [30], the main reported side effect, and the mechanism of action of these treatments require confirmation of the amyloid pathology and its relationship with the objective cognitive decline. A comprehensive diagnostic approach is therefore essential to consider the potential of these drugs to modify the course of the disease. In response to these advancements, the French federation of memory clinics (Fédération des Centres Mémoire [FCM]) formed a working group comprising neurologists, geriatricians, and general practitioners to develop updated recommendations for diagnosing AD. This effort aimed to incorporate recent diagnostic advances and to provide a standardized approach to improve early detection and management.

2. Methods

The FCM represents all memory clinics across France, including tertiary university memory clinics, hospital memory clinics, and private practice. The FCM is a scientific society dedicated to formulating clinical practice guidelines for the management of neurocognitive diseases.

In response to the potential emergence of new therapeutic strategies targeting the early phase of AD, the FCM initiated the development of updated recommendations for the diagnosis of early AD by establishing an expert panel comprising 11 specialists, including neurologists, geriatricians, and general practitioners with extensive experience in Alzheimer's disease management. The composition of this group was justified by the skills of practitioners involved in

each step of the patient pathway, from detection to diagnosis, and represented various memory clinic settings, including secondary and tertiary memory clinics, hospital and university clinics, private practice, as well as a general practitioner from the French National College of General Practitioners.

The working group adopted the clinical practice recommendation methodology proposed by the French National Authority for Health (HAS) [31]. Initially, the group collaboratively identified and selected relevant questions through consensus. These questions were then addressed by subgroups, who conducted comprehensive literature reviews to gather pertinent data. Monthly meetings were held for subgroups to present their findings and proposed recommendations based on the literature review and expert consensus. Discussions focused on the integration of scientific data and their translation into practical clinical recommendations based on published results and literature reviews, with a target of at least 75% agreement among experts in face-to-face collegial meetings before finalization of the recommendations.

Following the drafting of the initial recommendations, an external review was conducted to ensure the validity and applicability of the proposed guidelines. Twelve external reviewers, including neurologists, geriatricians, and psychiatrists with significant experience in AD management and diagnosis, as well as two general practitioners, were involved in this process. Their feedback was carefully considered and incorporated into the final version of the recommendations.

A comprehensive and detailed version of these recommendations in French can be found on the FCM website (<https://www.centres-memoire.fr/>). This article provides a condensed version, summarizing the essential key points of these recommendations.

3. Recommendations

3.1. What are the different stages of AD?

The working group proposes using the following symptomatic clinical stages for AD, which are largely inspired by the clinical staging proposed by the Alzheimer's Association Workgroup in 2024 [27], considering only symptomatic stages, i.e., those characterized by alterations on objective cognitive tests:

- minor neurocognitive disorder due to AD:
 - cognitive performance is impaired/abnormal based on objective, standardized cognitive tests,
 - evidence of decline from baseline, documented through the individual's or informant's interview (e.g., partner) or by changes in longitudinal cognitive tests or behavioral assessments,
 - ability to perform activities of daily living independently, although cognitive difficulties may have a detectable functional impact on complex daily activities, i.e., they may take more time or be less efficient but are still completed, as reported by the patient and/or corroborated by the informant;
- major neurocognitive disorder with mild functional impairment (mild stage):

- progressive cognitive decline and mild functional changes in instrumental activities of daily living (IADLs) such as meal preparation, household tasks, using transportation, and managing finances, while maintaining independence in basic activities of daily living (BADLs) such as bathing, dressing, transferring, and eating prepared meals;
- major neurocognitive disorder with moderate functional impairment (moderate stage):
 - progressive cognitive decline and moderate functional changes in BADLs requiring assistance,
- major neurocognitive disorder with severe functional impairment (severe stage):
 - progressive cognitive decline and severe functional changes, with total dependency on others for BADLs.

The working group emphasizes that:

- assessing the clinical stage of AD can be complex in the early stages and requires the expertise of physicians trained in memory consultation;
- in routine clinical practice, the disease stages can be assessed through detailed interviews evaluating the impact on autonomy in daily life (BADLs and IADLs) [32], and dependency on aids;
- interviewing a close family informant who spends several hours a week with the patient is particularly important for assessing the stage of the disease;
- in a clinical research context, administering the clinical dementia rating (CDR) scale [33] allows for a more refined assessment of the stage of AD.

3.2. What is the definition of early diagnosis for AD?

The diagnosis of AD at an early stage is a clinico-biological diagnosis, requiring both evidence of abnormal cognitive performance on neuropsychological assessment and positive biomarkers indicating beta-amyloid pathology (decreased A β 42 or A β 42/40 ratio in cerebrospinal fluid [CSF] or positive amyloid PET) and phosphorylated tau protein accumulation (increased phosphorylated tau in CSF).

A neuropsychological evaluation is needed to diagnose AD at an early stage [34]. Indeed, objective cognitive impairment may be subtle at the stage of minor neurocognitive disorder and requires specific testing, particularly for verbal episodic memory. Neuropsychological assessment is essential to determine the phenotypic profile of AD. Impairment in verbal episodic memory, especially a storage deficit that is poorly responsive to cueing [35,36], is particularly suggestive of the diagnosis. Other common AD phenotypes include the logopenic variant of primary progressive aphasia, and posterior cortical atrophy [25]. In contrast, uncommon phenotypes such as the behavioral or dysexecutive variant, corticobasal syndrome, the non-fluent variant of primary progressive aphasia, and the semantic variant [25] should prompt a careful re-evaluation of the diagnosis, especially in cases where biomarker profiles are discordant or atypical.

Early-stage diagnosis requires the expertise of a physician trained in memory consultations, integrating all clinical and paraclinical data.

The diagnosis is considered early if made at the stage of minor neurocognitive disorder or major neurocognitive disorder with mild functional impairment. Isolated cognitive complaints in the absence of indicative findings in the neuropsychological assessment do not support an early AD diagnosis.

A diagnosis of minor neurocognitive disorder is made when the patient is still able to plan routes and manage administrative tasks independently. At this stage, overall cognitive efficiency is preserved, usually with a Mini-Mental State Examination (MMSE) score ≥ 27 and a global CDR score of 0.5, interpreted within the patient's socio-cultural context.

The diagnosis of major neurocognitive disorder with mild functional impairment is made when a patient experiences an impact on their daily life, such as managing administrative tasks or planning unfamiliar routes in relation to their cognitive deficits. At this stage, the MMSE score is usually between 21 and 26 [37], and the CDR score is 1, interpreted within the patient's socio-cultural context.

Specific biomarker evaluation is recommended for early diagnosis [15], provided there are no contraindications, as clinical symptoms alone have imperfect specificity for diagnosing Alzheimer's pathology. Lumbar puncture can assess both beta-amyloid (A β 42, A β 42/40 ratio) and phosphorylated tau proteins levels [12,16]. If lumbar puncture is contraindicated or impossible, amyloid PET may be proposed as an alternative assay.

If biomarker analysis is not possible, the diagnosis of AD can be partially supported by longitudinal follow-up and the absence of an alternative diagnosis.

3.3. What is the role of early diagnosis of AD according to age?

Regardless of the patient's age, it is required to clearly understand the patient's expectations, concerns, and desire for a diagnosis. It is important to distinguish between the patient's wishes and those of their family.

In younger patients, the significant impact of their cognitive impairment on their professional and family life often results in a strong desire for an explanation and a precise diagnosis. To meet this need, it is essential to raise awareness among healthcare professionals (and the general public) about the clinical particularities of early-onset AD to enable the earliest possible diagnosis. This includes enhanced coordination with occupational health services, as these patients' difficulties are often first noticed in a professional context.

The young or very advanced age of patients adds an additional layer of complexity to the etiological diagnosis and prognosis of AD [38]. In young patients, the frequent delay in diagnosis is particularly detrimental to their proper care [39]. Conversely, although advanced age should not, in itself, preclude the diagnosis of AD, it should prompt careful consideration of the benefit-risk balance of complex additional investigations, taking into account the patient's context and expectations. Moreover, in older adults, co-pathologies (neurodegenerative and cerebrovascular lesions), frailty and comorbidities (psychiatric conditions, poly-pathology, poly-medication) contribute to clinical phenotypes and complicate accurate diagnosis [40]. In this context, the presence of a brain

pathology (e.g., amyloid pathology) does not necessarily mean that the phenotype is driven by this pathology (i.e., AD), as amyloid pathology is common in later life, and several other factors may contribute to the cognitive impairment [41].

3.4. Which tools should be used to detect neurocognitive disorders in primary care?

Although the issue of detecting cognitive disorders in primary care is crucial, there is currently no specific recommendation in France on the tools to be used in this context. Based on the literature and the participants' experience, and bearing in mind that clinical research is currently underway on this subject, the working group recommends the following tools for detecting cognitive disorders that may suggest AD in general practice, provided the recommended test conditions are carefully followed and referring to the normal values:

- MMSE score (essential) [42]: the MMSE is used as a test to assess global cognitive efficiency. Despite its limitations, the MMSE has become the minimal cognitive test for characterizing the patient's cognitive state and contributes to the assessment of future cognitive follow-up. Its administration time is usually less than 15 minutes, making it feasible for use in primary care;
- mini-IADL (essential) with assessment of four items (telephone, transport, budgeting, medication management) [43]: this tool is used to characterize the impact on daily life. It helps assess the patient's ability to perform complex instrumental activities of daily living;
- a verbal episodic memory test (if possible): the five-word test [44] or the memory impairment screen (MIS) can be used to assess episodic memory. These tests are brief and can provide valuable information about the patient's memory function;
- clock drawing test (if possible): the clock drawing test is a simple, quick assessment with relevant psychometric properties for detecting cognitive disorders in primary care. It can be used as an intermediate task alongside the five-word test, before the delayed recall part of this test.

The working group emphasizes the need for research in primary care to optimize detection strategies for cognitive disorders.

3.5. Which first-line paraclinical assessment should be performed in primary care for a patient suspected of neurocognitive disorder?

The working group recommends conducting a first-line assessment, which should include:

- biological assessment:
 - perform a complete blood count, C-reactive protein, blood electrolyte panel including sodium, calcium, glucose, albumin, urea, creatinine, and thyroid-stimulating hormone,
 - depending on clinical context: vitamin B12, folate levels, liver function tests, Lyme disease serology, HIV, and syphilis serologies,

- if these biological tests have been performed within the past 12 months as part of ongoing follow-up, there is no need to prescribe them again;
- brain imaging:
 - systematic brain imaging is recommended for any recent onset of confirmed cognitive impairment [45],
 - preferably, brain magnetic resonance imaging (MRI) should be performed (a computed tomography scan is reserved for MRI contraindications such as claustrophobia, non-MRI-compatible foreign material, patient agitation),
 - the brain MRI should include at least a volumetric T1 sequence, a FLAIR sequence, a diffusion-weighted imaging sequence, and a bleeding-sensitive sequence (T2*, SWI or other susceptibility-weighted magnetic resonance sequences),
 - contrast agent injection is not necessary for the first-line assessment;
- the objectives of brain imaging include [46]:
 - to rule out other potential causes (e.g., intracranial mass, chronic adult hydrocephalus, stroke sequelae, subdural hematoma),
 - to identify associated vascular lesions (e.g., white matter hyperintensities related to microangiopathy quantified according to the Fazekas scale, microbleeds, lacunar infarcts and enlarged perivascular spaces),
 - to quantify any potential hippocampal atrophy (using the Scheltens medial temporal atrophy scale or automated measurements) and any other focal or global atrophy.

3.6. Which patients should be referred to a memory clinic?

All patients with objectively identified neurocognitive disorders can be referred to a memory clinic to confirm the syndromic diagnosis, to establish an etiological diagnosis, and to propose a personalized care plan, particularly to address aggravating and modifiable factors [47]. This personalized care plan will be adaptive and should be re-evaluated in coordination with primary care.

A memory consultation is particularly recommended in cases of patient and/or family concern, early-stage disease, atypical (i.e., non-amnesic) clinical presentation, patients with age of onset before 65 or who are still professionally active, and those with first-degree family history of neurocognitive disorders.

The nature and framework of the memory consultation depend on local contexts and organizations, as well as individual situations. The appropriateness of a memory consultation should be discussed with the patient, taking into account their perspectives, desires, and fears about the diagnosis.

Situations can be complex when clinicians and/or family members identify risky behaviors (e.g., driving, financial mismanagement, medication errors) in a patient who appears to be anosognosic or in denial of the disease. In such cases, the memory consultation can help the patient gain partial awareness of their difficulties and facilitate care. However, it should be noted that some measures should not wait for the specialized memory consultation but can be initiated immediately in primary care (e.g., driving advice, legal protection

measures, nursing care prescriptions for medication management).

Finally, we recommend a graduated and personalized diagnostic strategy. This strategy includes initial detection, evaluation of cognitive complaints, syndromic diagnosis (isolated complaints, minor neurocognitive disorder, major neurocognitive disorder), and the prescription of additional tests (brain imaging and blood tests) in primary care prior to referral to a memory consultation.

3.7. What is the role of neuropsychological assessment?

Neuropsychological assessment is central to the diagnostic process for neurocognitive disorders [48]. It is indicated for patients presenting with suspected cognitive impairment following a memory consultation that includes a preliminary screening.

The neuropsychological assessment allows for:

- objectifying cognitive deficits in response to a complaint;
- determining the intensity/severity of cognitive impairments;
- identifying the affected and preserved cognitive domains: episodic memory, attentional and executive functions, instrumental functions (language, visuo-perceptive and visuospatial functions, praxis), which helps establish the cognitive profile and guide the diagnosis;
- assessing anxiety levels, mood, and other possible comorbidities;
- guiding rehabilitation efforts and adapting the implementation of support at home;
- providing quantitative evaluation for patient monitoring.

However, in some situations, neuropsychological assessment is either not indicated or should be deferred:

- severe and uncontrolled psychiatric disorders;
- ongoing alcohol or other psychoactive substance abuse;
- advanced cognitive impairment, typically with an MMSE score ≤ 15 , depending on the clinical context, the nature of the impairments (e.g., language impairment affecting the MMSE), and the patient's education level.

The proposed assessment must consider the patient's education level, and an assessment adapted to illiteracy should be offered to patients with less than five years of schooling or who do not master reading and writing. The presence of an interpreter, external to the family, should be considered for patients who do not speak French.

3.8. What are the indications for assessing AD biomarkers in CSF?

Considering the current diagnostic performance of biomarkers and their importance in routine care, the working group recommends that:

- the decision to measure AD biomarkers in the CSF should result from a dialog between a physician trained in memory consultation and the patient, or if appropriate, the patient's

family. The patient should be informed and able to understand the diagnostic and therapeutic implications and express their consent for the procedure;

- biomarker measurement in the CSF can be proposed for any patient with major neurocognitive disorder without a formally identified cause to support or rule out the diagnosis of AD;
- biomarker measurement can be proposed to a patient with minor neurocognitive disorder, particularly in the case of documented impairment in verbal episodic memory using an objective test, after the diagnostic and therapeutic implications have been explained;
- measuring CSF biomarkers appears appropriate, in particular for patients with an age of onset younger than 66 years old, in cases of minor cognitive impairment suggestive of early-stage disease eligible for a specific therapy, and in suspected non-amnesic forms of AD (dysexecutive, language, or visuospatial forms);
- based on current knowledge, it is not recommended to measure CSF biomarkers in patients with cognitive complaints without objective abnormalities on neuropsychological assays or in asymptomatic individuals.

3.9. What is the role of blood-based biomarkers in the diagnosis of AD?

Currently, blood-based biomarkers for AD do not have a validated role in routine care, whether in primary care or specialized memory consultations.

Recent studies have shown that the measurement of phosphorylated tau protein in plasma (in particularly p-tau 217) appears particularly promising for identifying patients with cerebral amyloidopathy and could be integrated into patient stratification strategies in the future [49,50].

Ongoing research will further clarify how blood-based biomarkers might be integrated into detection and diagnostic strategies, potentially leading to an update of these recommendations in the future.

3.10. What is the contribution of FDG-PET in the diagnosis of AD?

[18F]-fluorodeoxyglucose positron emission tomography (FDG-PET) brain imaging can be performed after neuropsychological assessment in patients presenting with objective cognitive impairment [51,52]. It can help identify focal cortical hypometabolism, the topography of which can guide etiological diagnosis (e.g., hypometabolism of posterior associative regions in AD, hypometabolism of the visual cortex, or the posterior cingulate island sign in dementia with Lewy bodies):

- its indication should be discussed by a nuclear medicine physician and a physician expert in the care of neurocognitive pathologies;
- FDG-PET brain imaging is particularly indicated when:
 - the patient presents cognitive impairments suggestive of a non-amnesic presentation of AD,
 - the patient presents cognitive impairments suggestive of a neurodegenerative disease other than AD, such as suspected frontotemporal lobar degeneration, LATE

pathology, dementia with Lewy bodies, or cerebrovascular diseases,

- the patient presents cognitive impairments suggestive of AD, but CSF biomarkers are contraindicated, negative, or discordant, and/or amyloid PET is not accessible, not interpretable, or negative,
- the clinical presentation is atypical or inconsistent raising doubts about the organicity of the symptoms (suspicion of functional cognitive disorder or suspicion of a psychiatric disorder);
- FDG-PET brain imaging is usually not indicated in typical (amnesic) cases of AD presentation with positive CSF biomarkers or amyloid PET.

3.11. What are the indications for amyloid PET in the diagnosis of AD?

Amyloid PET brain imaging may be indicated for patients presenting with major or minor neurocognitive disorders suspected to be secondary to AD [53], following a comprehensive neuropsychological evaluation, in cases of an atypical clinical presentation (including early onset of symptoms before the age 65), associated with a contraindicated lumbar puncture or when CSF biomarker results are discordant or inconclusive.

The indication of this examination should be discussed and validated within an interdisciplinary framework involving a nuclear medicine physician (specifically trained in the amyloid-PET procedures) and an expert physician practicing in a secondary or tertiary memory clinic. This validation aims to ensure the absence of diagnostic alternatives and the appropriateness of the indication for the patient's clinical situation.

3.12. What are the indications for genetic testing?

For early-onset AD, genetic testing for pathogenic mutations should be conducted in accordance with the guidelines set forth by the National Reference Center for Young-Onset Alzheimer's Disease (CNR-MAJ) [54]. Healthcare providers must ensure that patients receive comprehensive counseling regarding the implications and potential consequences of genetic analysis prior to testing.

Genetic testing should be proposed to any patient developing AD (confirmed by CSF biomarkers or amyloid PET) before the age of 66, even in the absence of other family history.

In patients with late-onset AD (onset after the age of 65), the current state of knowledge does not justify diagnostic genetic analysis, with rare exceptions.

In the future, the use of APOE genotype analysis may be recommended if it is integrated into a decision-making algorithm for initiating or monitoring treatment with anti-amyloid immunotherapy.

3.13. What are the modalities for disclosing an AD diagnosis?

The physician should ensure, as much as possible, that the patient wishes to be informed of the AD diagnosis [55]. If the patient wishes to be informed, the diagnosis should be

communicated as precisely as possible by the physician who conducted the evaluation and made the diagnosis. When uncertainty persists, a syndromic diagnosis can be proposed with multiple etiological diagnostic hypotheses, or a probabilistic diagnosis can be formulated. This diagnosis should be included in correspondence with other healthcare providers to ensure consistent communication. The physician must respect the wishes of patients who do not want to be informed of their diagnosis.

The presence of a close caregiver is useful to ensure that the patient understands the diagnosis, but this should only occur with the patient's consent. In cases of severe cognitive impairment where the patient's consent cannot be obtained, the diagnosis may be communicated to the caregiver.

The diagnostic disclosure should be documented in a summary letter addressed to all medical correspondents involved in the patient's care. This letter should include at least a syndromic diagnosis, an etiological diagnosis if possible, detailing any doubts or formulating multiple hypotheses ranked according to their probability. If necessary, the prescribed pharmacological and non-pharmacological treatments, the recommendations for administrative procedures, and the proposed follow-up modalities should also be included.

Certain principles must be respected: the disclosure should take place in a calm environment, with dedicated time for explanation and discussion. Associated depression should be identified and treated. Neuropsychological battery results should be disclosed and explained, with the emphasis on the preserved cognitive function. The explanation should be adapted to the patient's and their family's understanding capacity. The name of the disease should be clearly stated after explaining what it refers to and the diagnostic reasoning. The referring physician should address any comorbid depression, provide reassurance by emphasizing preserved abilities and the slow progression of symptoms, offer a care plan and follow-up, and inform about available support resources, therapeutic education, and potential research participation.

3.14. What is the role of a post-diagnostic disclosure consultation?

The post-diagnostic disclosure consultation aims to [56]:

- evaluate the psychological impact of the diagnosis disclosure;
- review and sometimes rephrase what was discussed during the initial disclosure;
- facilitate meetings with other professionals in a multidisciplinary consultation setting (psychologist, consultation nurse, social worker, etc.);
- discuss potential genetic analysis if indicated and any available clinical research if it was not covered during the initial diagnostic disclosure consultation.

This consultation should be prioritized in complex situations (e.g., extreme ages, isolated patients, psychological distress following diagnosis disclosure, or comorbid conditions of the patient and/or the caregiver).

4. Discussion

The FCM has developed a set of diagnostic recommendations that reflect the evolution of AD from a clinical syndrome to a clinico-biological entity with identifiable stages and therapeutic implications. Building on the IWG-2 framework [25,26], the FCM defines AD as a symptomatic, clinico-biological condition, stressing early identification through cognitive testing and validated biomarkers. This orientation aligns with the therapeutic imperative to intervene as early as possible, particularly given the efficacy profile of emerging anti-amyloid immunotherapies.

Several key points are emphasized in these recommendations. Notably, AD diagnosis requires both clinical symptoms suggestive of the disease on neuropsychological examination and positive biomarkers for beta-amyloid peptide and phosphorylated tau. We emphasize here the crucial importance of detecting cognitive disorders in primary care, and the importance of not trivializing a cognitive complaint or deficit, whatever the patient's age. This involves raising awareness among healthcare professionals, first and foremost general practitioners, as well as the general public. The patient plays a central role in the diagnostic approach: they must understand the implications and give their consent to initiate an etiological investigation for neurocognitive disorders, particularly for biomarker evaluation.

These FCM guidelines bridge biological precision with real-world application. CSF biomarker testing (A β 42/40, p-tau) is recommended early in the diagnostic process, particularly in cases where the clinical phenotype is compatible with AD. While biomarker assessment may be considered in selected uncommon phenotypes, it should be interpreted cautiously and always within the broader clinical context to avoid misattribution of Alzheimer's pathology. FDG-PET and amyloid PET are used selectively, depending on biomarker availability, atypical presentation, or discordant findings. This tiered approach maximizes diagnostic accuracy while considering system-level constraints. It is important to note that these are French recommendations, primarily intended for use within France. The authors acknowledge that there is considerable geographical heterogeneity in certain clinical practices, particularly regarding the use of AD biomarkers. For instance, in France, amyloid PET imaging is considered as a second-line examination, indicated in cases where CSF biomarkers are contraindicated or yield contradictory results.

The field of AD is rapidly evolving, with numerous conceptual changes occurring frequently. As such, these recommendations are expected to evolve over time, in line with advancing knowledge and changing practices. In particular, the role of blood-based biomarkers, the expanding use of anti-amyloid immunotherapies, and the question of AD screening are areas that may see significant developments in the near future.

These recommendations provide a precise, scalable, and ethically structured framework for standardizing and harmonizing the diagnostic approach to AD in France, while recognizing the need for flexibility in their application. They aim to improve early diagnosis, facilitate timely interventions, and enhance patient care. However, as with any guidelines,

they should be applied in conjunction with clinical judgment and individualized patient considerations. The authors emphasize the importance of continued research and international collaboration to further refine our understanding of AD and improve diagnostic and therapeutic strategies. As new evidence emerges, these recommendations will be reviewed and updated to ensure they reflect the most current and evidence-based practices in the field of AD diagnosis and management.

Disclosure of interest

Julien Dumurgier has received personal fees for lectures or advice from: Lilly, Eisai, Novo Nordisk, Roche. He has been investigator for: Eisai, Lilly, Novo Nordisk, Roche, Alzheon. Antoine Garnier-Crussard is an unpaid sub-investigator or local principal investigator in clinical trials and studies funded by UCB Pharma, Biogen, TauRx Therapeutics, Roche, Novo Nordisk, Alzheon, Medesis Pharma, GlaxoSmithKline, Eisai, Lilly, MSD Avenir. Isabelle Rouch benefited from hospitality funding from Biogen for scientific meetings. Marc Verny has received personal fees for lectures or advice from: Eisai and Lilly. David Wallon is an unpaid investigator in clinical trials and studies funded by Novartis, Roche, Alzheon, BMS, Lilly.

The other authors declare that they have no competing interest.

Acknowledgments

The authors would like to thank the members of executive group leading the French Federation of Memory Clinics (FCM) for their valuable input throughout the development of the recommendations. We also thank the external reviewers for their critical feedback and contributions, namely: Stéphanie Bombois (Paris), Vincent Bultingaire (Dignes-les-Bains), Leslie Cartz-Piver (Limoges), Matthieu Ceccaldi (Marseille), Julien Delrieu (Toulouse), Jean-Michel Dorey (Lyon), Eric Dumas (Périgueux), Virginie Goutte (Saint-Cloud), Thibaud Lebouvier (Lille), Philippe Marissal (Champagne-en-Valromey), Céline Petitgas (Paris), Maria Soto-Martin (Toulouse), Valérie Tcherniack (Marseille), Mélissa Tholomier (Arceuil), and Thomas Tannou (Montreal).

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